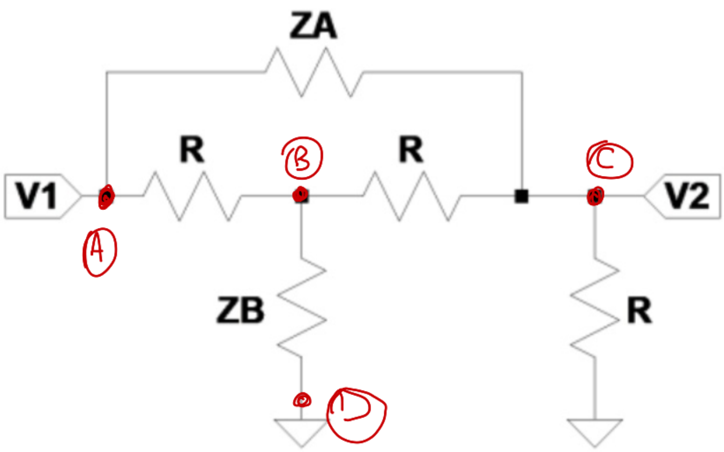
Ejercicio #5

Dado el siguiente cuadripolo cargado con R y cumpliendo con la condición de diseño $Z_A . Z_B = R^2$ La red se conoce como ***Red de Resistencia constante***.

Aplique MAI para obtener $\frac{V_2}{V_1}$ y Z_{in}



Obtengo parámetros MAI por definición

$$y_{mn} = \left. \frac{I_m}{V_n} \right|_{V_k = 0 ,\ k \neq n}$$

y aprovechando propiedad de Sumatoria NULA

$$Y_{MAI} = \begin{array}{c} \\ A \\ B \\ C \\ D \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{pmatrix} \frac{1}{Z_A} + \frac{1}{R} & -\frac{1}{R} & -\frac{1}{Z_A} & 0 \\ -\frac{1}{R} & \frac{2}{R} + \frac{1}{Z_B} & -\frac{1}{R} & -\frac{1}{Z_B} \\ -\frac{1}{Z_A} & -\frac{1}{R} & \frac{2}{R} + \frac{1}{Z_A} & -\frac{1}{R} \\ 0 & -\frac{1}{Z_B} & -\frac{1}{R} & \frac{1}{Z_B} + \frac{1}{R} \end{pmatrix} \end{array}$$

$$\frac{V_2}{V_1} = \frac{V_{CD}}{V_{AD}}$$

$$A_{AD}^{CD} = \operatorname{Sgn}(C-D) \cdot \operatorname{Sgn}(A-D) \cdot \frac{\underline{Y}_{CD}^{AD}}{\underline{Y}_{AD}^{AD}}$$

$$\underline{Y}_{CD}^{AD} = \begin{vmatrix} -\frac{1}{R} & \frac{2}{R} + \frac{1}{ZB} \\ -\frac{1}{ZA} & -\frac{1}{R} \end{vmatrix} = \frac{1}{R^2} + \frac{2}{R \cdot ZA} + \frac{1}{ZAZB}$$

$$= \frac{1}{R^2} + \frac{2ZB + R}{R \cdot ZA \cdot ZB} = \frac{ZA \cdot ZB + 2ZBR + R^2}{R^2 ZA ZB}$$

$$= \frac{R^2 + 2ZBR + R^2}{R^4} = \frac{2R + 2ZB}{R3}$$

$$\underline{Y}_{AD}^{AD} = \begin{vmatrix} \frac{2}{R} + \frac{1}{ZB} & -\frac{1}{R} \\ -\frac{1}{R} & \frac{2}{R} + \frac{1}{ZA} \end{vmatrix} = \frac{4}{R^2} + \frac{2}{RZA} + \frac{2}{RZB} + \frac{1}{ZAZB} - \frac{1}{R^2}$$

$$= \frac{3}{R^2} + \frac{2ZA + 2ZB}{RZAZB} + \frac{1}{ZAZB}$$

$$= \frac{4}{R^2} + \frac{2ZA + ZB \cdot 2}{R^3} = \frac{4R + 2ZA + 2ZB}{R^3}$$

$$A = \frac{2R + 2 \cdot ZB}{4R + 2ZA + 2ZB}$$

$$ZB = \frac{R^2}{ZA}$$

$$\boxed{A = \frac{R + ZB}{2R + ZA + ZB} = \frac{R \cdot \left(1 + \frac{R}{ZA}\right)}{R \cdot \left(2 + \frac{ZA}{R} + \frac{R}{ZA}\right)}}$$

Para Zin

$$Z_{mn} = \frac{\underline{Y}^{mn}_{mn}}{\underline{Y}^{m}_{n}}$$

$$Z_{in} = Z_{AD} = Z_{14} = \frac{\underline{Y}^{AD}_{AD}}{\underline{Y}^{A}_{D}}$$

$$Y^{AD}_{AD} = \frac{4R + 2ZA + 2ZB}{R^3}$$

$$Y_{AD} = Y_{CB} = (-1)^{3+4} \begin{vmatrix} \frac{1}{ZA} + \frac{1}{R} & -\frac{1}{ZA} & 0 \\ -\frac{1}{R} & -\frac{1}{R} & -\frac{1}{ZB} \\ 0 & -\frac{1}{R} & \frac{1}{ZB} + \frac{1}{R} \end{vmatrix}$$

$$= -1 \cdot \left(\frac{1}{Z_A} + \frac{1}{R}\right) \cdot \left(-\frac{1}{R}\right)\left(\frac{1}{Z_B} + \frac{1}{R}\right) - \left[\frac{1}{R} \cdot \frac{1}{Z_A} \cdot \left(\frac{1}{Z_B} + \frac{1}{R}\right) + \frac{1}{R}\frac{1}{Z_B}\left(\frac{1}{Z_A} + \right.\right.$$

$$= -1 \cdot \left(-\frac{1}{Z_A R} - \frac{1}{R^2}\right) \cdot \left(\frac{1}{Z_B} + \frac{1}{R}\right) - \left[\frac{1}{R Z_A Z_B} + \frac{1}{R^2 Z_A} + \frac{1}{A Z_B Z_A} + \frac{1}{R^2 Z_B}\right]$$

$$= \frac{1}{Z_A Z_B R} + \frac{1}{Z_A R^2} + \frac{1}{Z_B R^2} + \frac{1}{R^3} + \frac{2}{R Z_A Z_B} + \frac{1}{R^2 Z_A} + \frac{1}{R^2 Z_B}$$

$$= \frac{3}{Z_A Z_B R} + \frac{2}{Z_A R^2} + \frac{2}{Z_B R^2} + \frac{1}{R^3}$$

$$= \frac{4}{R^3} + \frac{2}{Z_A R^2} + \frac{2}{Z_B R^2} = \frac{4 Z_A Z_B + 2 Z_A R + 2 Z_B R}{R^3 Z_A Z_B}$$

$$= \frac{4R^2 + 2 Z_A R + 2 Z_B R}{R^5} = \frac{4R + 2Z_A + 2Z_B}{R^4}$$

$$Z_{in} = \frac{\cancel{\dfrac{4R + 2Z_A + 2Z_B}{R^3}}}{\cancel{\dfrac{4R + 2Z_A + 2Z_B}{R^4}}} = \boxed{R}$$